

Mathematical Principles for Understanding Transformer-Based Models

1. Probability & Distributions

Softmax: $P(y_i) = \exp(z_i) / \sum_j \exp(z_j)$. Converts logits into probabilities, which express confidence in outputs. Cross-Entropy Loss: $L = -\sum y_i \log(\hat{y}_i)$. The training objective, showing models optimize to match text patterns, not truth.

2. Linear Algebra

Dot Product: $\text{score}(q, k) = q \cdot k$. Measures alignment between query and key vectors. Cosine Similarity: $\cos(\theta) = (A \cdot B) / (||A|| ||B||)$. Compares embeddings (semantic similarity). Matrix Multiplication: Core operation in each transformer layer.

3. Self-Attention Mechanism

Attention(Q, K, V) = $\text{softmax}(QK^T / \sqrt{d_k}) V$. Lets tokens attend to one another. Multi-Head Attention combines multiple perspectives, e.g., one head might attend to harm, another to duty.

4. Nonlinearities & Representations

Feedforward Layers: $f(x) = W_2 \sigma(W_1 x + b_1) + b_2$. Builds abstract features beyond token-level. Embeddings: Map discrete tokens into high-dimensional vectors; similar meanings cluster together.

5. Uncertainty & Calibration

Brier Score: $(1/N) \sum (p_i - y_i)^2$. Measures accuracy of probability estimates. Expected Calibration Error (ECE): $\sum_m |acc(B_m) - conf(B_m)|$. Tests whether confidence aligns with actual accuracy.

6. Optimization & Scaling

Gradient Descent: $\theta \leftarrow \theta - \eta \nabla_{\theta} L$. Parameters updated to reduce loss. Scaling Laws: $\text{Loss} \approx N^{-\alpha}$. Larger models trained on more data predictably improve performance, which may affect moral behavior artifacts.

Minimal Takeaway:

- Softmax = confidence distribution.
- Cross-entropy = pattern-matching objective.
- Attention = how the model decides what matters.
- Cosine similarity = grouping of moral concepts.
- Calibration = whether the model knows when it doesn't know.